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**Welcome
To
Network for you
CoPP**



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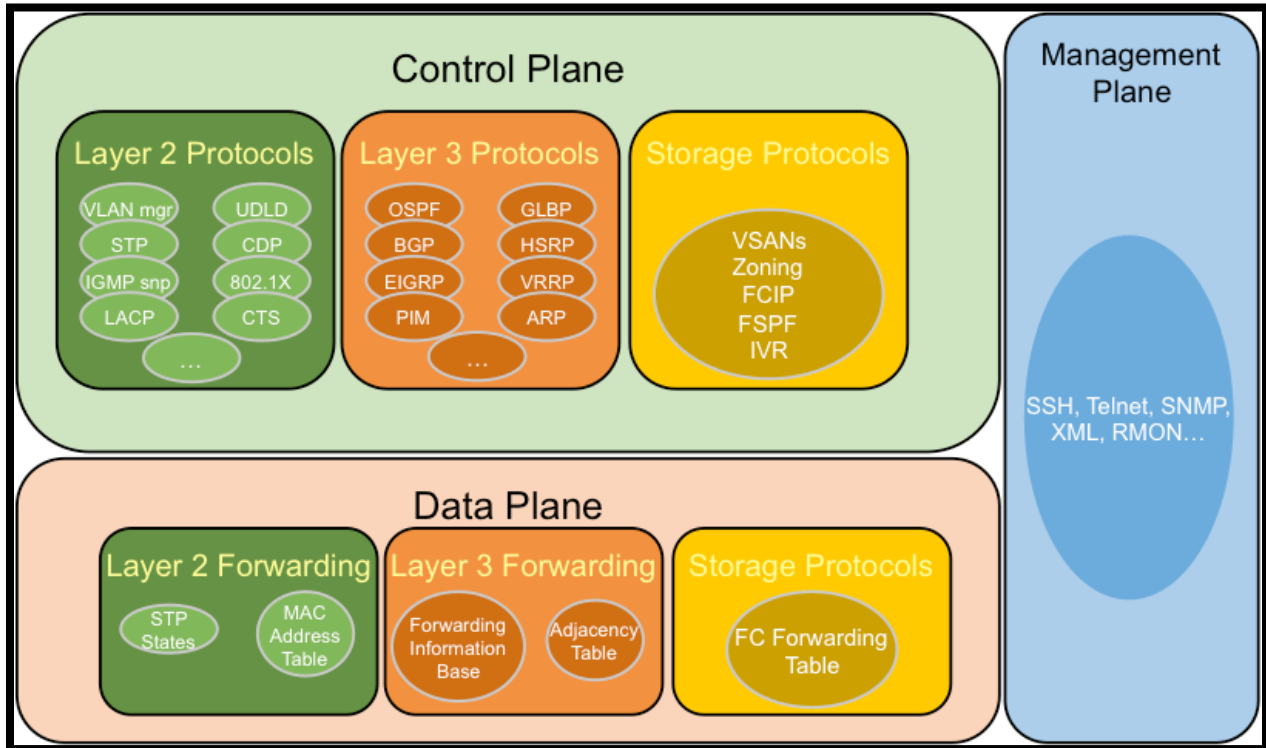
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Control Plane Policing (CoPP): Allows you rate-limit traffic that goes to the route processor.

Before going to this we need to have Router/ Switch plan idea.

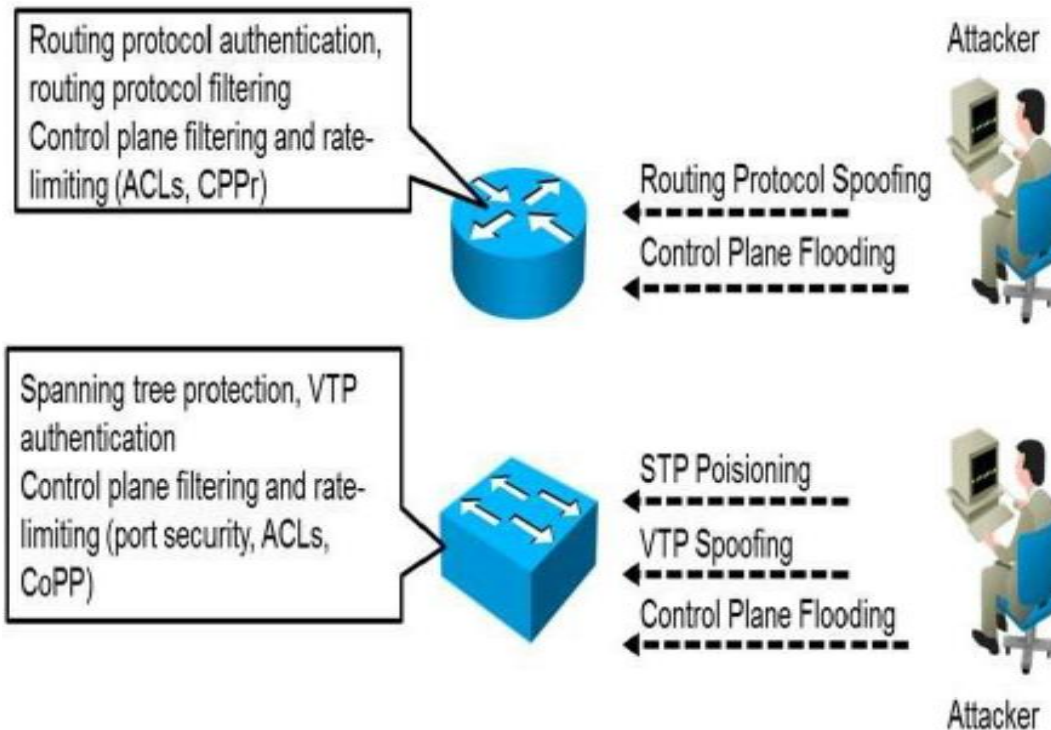


Control Plane: Control Plan traffic is from the **device to the device**. The control plane is responsible for exchanging routing information building the arp table etc. As given below the responsible of control plan.

- Learning MAC address to build a switch MAC address table
- Running STP to create a loop free topology.
- Building ARP tables
- Running routing protocols like OSPF, BGP, EIGRP etc. to build routing table.



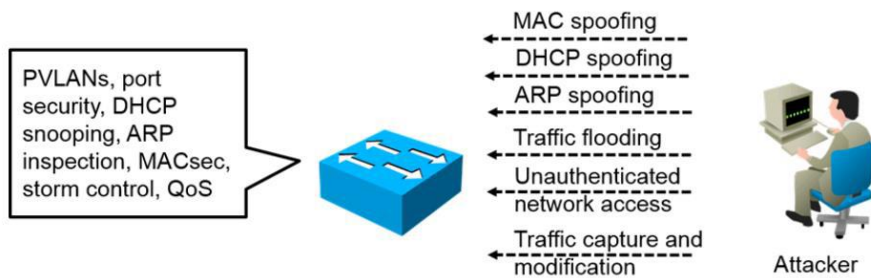
Control Plane Security Controls



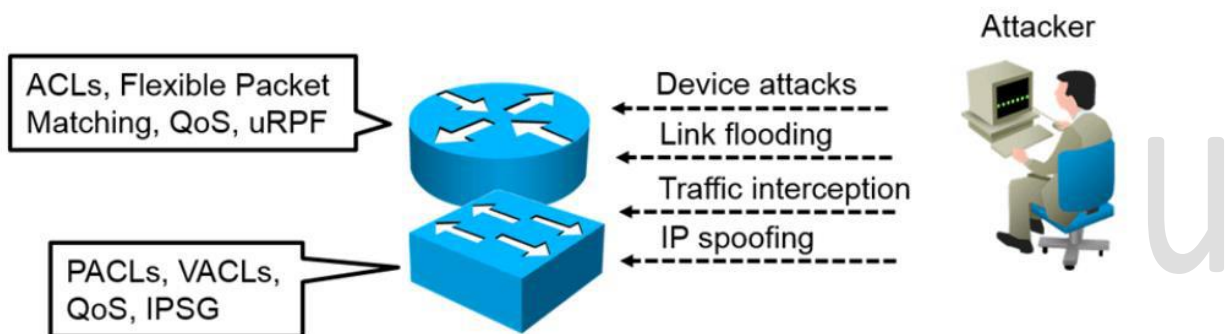
Data Plane: Data Plan traffic from the **user to the user**. Data plane is responsible for forwarding traffic and it relies on the information that gets from control plan. Here is some task that take care by Data Plan is given below.



Layer 2 Data Plane Security Controls



L3 Data Plane Security Controls



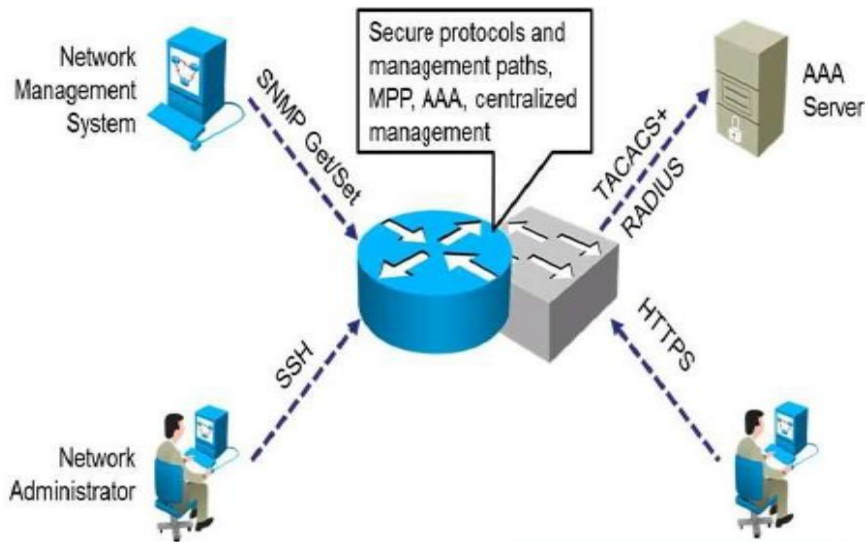
Management Plane:

Management Plane traffic is from the **user to the device**. Is used for access and management of our network devices. Management Plane protocols are FTP, HTTP, HTTPS, SSH, SNMP, Telnet, TFTP etc.

- Telnet
- SSH
- Console



Management Plane Security Controls



Best Practices for Securing the Management Plane.

- Enforce password policy on devices.
- Maximum number of login attempts, minimum password length.
- Custom privilege level assignments to restrict users.
- Use AAA services to authenticate, authorize & audit any commands.
- Keep accurate time using secure Network Time Protocol (NTP).
- Use encrypted and authenticated versions of SNMP.
- Encrypted management protocols such as SSH, HTTPs, SNMPv3.
- Enable Logging and monitoring on all network devices.
- Disable any unnecessary services on network devices.

Best Practices for Protecting the Data Plane.

- Create and enable Access Control List on network devices.
- Create and deploy Private VLAN (Virtual Local Area Network).
- Enable Spanning Tree guard such as BPDU Guard and Root Guard.



- Port security to protect against MAC address flooding and CAM overflow attacks.
- Enable DHCP snooping feature to prevent a rogue DHCP attack in the network.
- Dynamic ARP inspection (DAI) protect against ARP spoofing, ARP poisoning.

Best Practices for Protecting the Control Plane:

- Create and enable Control Plane Policing (CoPP) on devices.
- Identify type & rate of traffic reaches the control plane of Cisco IOS device.
- Allow users to manage the flow of traffic handled by the route processor.
- Prevent unnecessary traffic from overwhelming the route processor.
- Create and enable CPPr (Control Plane Protection) on network devices.
- Create and enable Port-filtering feature on network devices.
- Limits the number of packets for a specific protocol.
- Securing Routing Protocols use password authentication method.

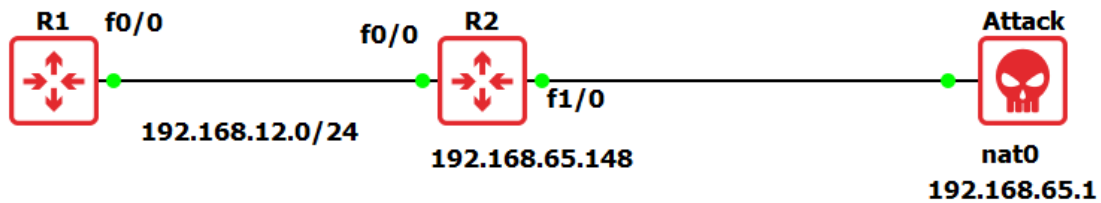
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Name	Use for
Create ACL	Identify the Traffic create Access Control List
Class Map	Identify the Traffic Use already created ACL
Policy Map	Policy Map take Action on Class Map
Control Plan	Apply the Policy Map using Service Policy

Lab Time:



Control Plan Policy Stop UDP DoS Attack



R1 Basic Configuration	R2 Basic Configuration
<pre>en config t hostname R1 int f0/0 ip add 192.168.12.1 255.255.255.0 no sh router ospf 1 int f0/0 ip ospf 1 area 0</pre>	<pre>en config t hostname R2 int f0/0 ip add 192.168.12.2 255.255.255.0 no sh int f1/0 ip add dhcp no sh router ospf 1 int f0/0 ip ospf 1 area 0</pre>



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UDP Unicorn

File View Tools Options Help

Target:

Port:

Packet Size: 50.0 KB

☐ Random Size

Delay: ms

Threads:

Sockets per Thread:

Data Sent: KB

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UDP Unicorn

File View Tools Options Help

Target: Port:

Packet Size: 50.0 KB
☐ Random Size

Delay: ms

Threads:

Sockets per Thread:

Data Sent: KB

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Control Plan Policy Stop UDP DoS Attack

```
R2(config)#ip access-list extended test
R2(config-ext-nacl)#permit udp any any
R2(config)#class-map cmap
R2(config-cmap)#match access-group name test
R2(config)#policy-map pmap
R2(config-pmap)#class cmap
R2(config-pmap-c)#police 8000 conform-action transmit exceed-action drop
R2(config-pmap-c-police)#exit
R2(config-pmap-c)#exit
R2(config-pmap)#exit
R2(config)#control-plane
R2(config-cp)#service-policy input pmap
R2#show policy-map control-plane
```

We can in R2:

```
R2#show policy-map control-plane
Control Plane

Service-policy input: pmap

Class-map: cmap (match-all)
 3566142 packets, 5369177227 bytes
 5 minute offered rate 63676000 bps, drop rate 63670000 bps
Match: access-group name test
police:
  cir 8000 bps, bc 1500 bytes
  conformed 471 packets, 547869 bytes; actions:
    transmit
  exceeded 3565671 packets, 5368629358 bytes; actions:
    drop
  conformed 8000 bps, exceeded 63670000 bps

Class-map: class-default (match-any)
 150 packets, 13760 bytes
 5 minute offered rate 0000 bps, drop rate 0000 bps
Match: any
```



Control Plan Policy Stop Ping Flood

```
R2(config)#ip access-list extended test
R2(config-ext-nacl)#permit icmp any any
R2(config)#class-map cmap
R2(config-cmap)#match access-group name test
R2(config)#policy-map pmap
R2(config-pmap)#class cmap
R2(config-pmap-c)#police 8000 conform-action transmit exceed-action drop
R2(config-pmap-c-police)#exit
R2(config-pmap-c)#log
R2(config-pmap-c)#exit
R2(config-pmap)#exit
R2(config)#control-plane
R2(config-cp)#service-policy input pmap
R2#show policy-map control-plane
```